

Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation

Abstract

Graph recommenders trained on logged data propagate preferences along walks that can traverse an edge more than once. With inverse-propensity or doubly robust (DR) edge estimates, such walks multiply an estimate with itself and bias multi-hop scores. We derive this bias exactly and remove it with DRUP, an idempotent walk correction that is exact for any odd number of hops and costs about as much as the uncorrected three-hop operator. We test it on Coat, Yahoo! R3, KuaiRand and KuaiRec (290 to 15,400 users; up to 10.3 million logged interactions) against up to 26 tuned baselines, and on a semi-synthetic KuaiRec log with known exposure. With true propensities and fixed weights, the correction reduces the error of the three-hop term by a factor of 4 to 12 and removes an 18–29% overestimate of an allocation’s utility; with estimated propensities, all operators overstate utility by 100–540%. Accuracy does not change: corrected and uncorrected three-hop DR propagation differ by at most 0.0014 nDCG. The guarantees need nuisances independent of the log: with degree weights computed from the log the correction no longer reduces bias, and the sample-split variant that satisfies them loses up to 0.06 nDCG on sparse logs. On KuaiRec, training-free DR propagation (nDCG@20 0.638) leads the best trained model by 0.004, within

that model’s seed spread.

Keywords: Causal recommendation, causal graph recommendation, doubly robust estimation, exposure bias, missing not at random, graph-based recommendation, exposure invariance.

1. Introduction

Graph-based recommenders represent users and items as nodes and logged interactions as edges, and they score a user–item pair by aggregating signals along multi-hop walks (He et al., 2020; Shen et al., 2021). The logged graph is not the preference graph. An edge exists only if the platform *exposed* the item and the user then responded, so the graph records the exposure policy as much as the user’s taste. Causal learning offers two families of corrections. *Loss-level* debiasing re-weights the training objective by inverse propensities or doubly robust (DR) terms (Schnabel et al., 2016; Wang et al., 2019; Saito et al., 2020), but it leaves the propagation operator of a graph model untouched. *Propagation-level* debiasing changes the propagated graph, for example by weighting each edge by its inverse propensity (Kim et al., 2022) or by learning popularity-aware aggregation weights (Zhou et al., 2023; Islam et al., 2026).

This paper starts from a simple observation about the second family. An inverse-propensity edge estimate $W_e = O_e Y_e / p_e$ is unbiased for the potential outcome Y_e , where O_e indicates that edge e was exposed, Y_e is the outcome and p_e the exposure propensity. A walk that traverses e twice contributes W_e^2 , and $\mathbb{E}[W_e^2] = Y_e / p_e \neq Y_e$. The same holds for DR edge estimates. Walks that repeat an edge are part of every multi-hop operator: the back-and-forth

step $u \rightarrow j \rightarrow u$ of a three-hop walk is one. Their excess is largest where the propensity is smallest, that is, on the rarely exposed edges that debiasing is meant to recover.

2. Research objectives and contributions

The aim of this study is to determine whether and when removing the repeated-walk bias helps graph recommendation from exposure-biased logs. We pursue four objectives, each tied to a research question (RQ).

- O1 (estimator).** Derive the exact bias of inverse-propensity and DR graph propagation and an estimator that removes it. *RQ1: how large is the bias, and can it be removed exactly at a cost comparable to the uncorrected operator?*
- O2 (conditions).** State the conditions under which the correction is unbiased and test them in the way practitioners fit nuisances. *RQ2: do the guarantees survive cross-fitting and data-dependent degree weights?*
- O3 (value).** Measure what the correction changes. *RQ3: does it improve the accuracy of recommendations against tuned baselines on four benchmarks, and does it improve the values of the scores on a log with known exposure?*
- O4 (invariance).** Test on real logs the prediction that expected scores do not depend on an item’s exposure. *RQ4: how informative is a thinning audit of this property?*

Answers in brief. The bias is exact and removable at three hops for the cost of one item Gram matrix (RQ1). The guarantees need nuisances that do not depend on the log: cross-fitting suffices if the degree weights are fixed a priori, but not if they are computed from the log, which our experiments do (RQ2). The correction leaves accuracy unchanged on four datasets, and it improves score values only under oracle conditions: true propensities and weights fixed in advance (RQ3). The thinning audit has little power and the selected configurations are not invariant on two of four datasets (RQ4).

Contributions.

1. *Diagnosis.* The exact conditional bias of inverse-propensity and DR propagation on unexposed candidates; under a propensity clip τ its worst case is $\Theta(1/\tau)$ (Theorem 5).
2. *Estimator.* DRUP (Doubly Robust, walk-Unbiased Propagation) uses $Y_e^2 = Y_e$ for binary outcomes and replaces each repeated edge by its idempotent unbiased counterpart. The three-hop operator has the asymptotic cost of the uncorrected one, and Möbius inversion over the coincidence patterns of walk indices makes the correction exact for any odd number of hops (Theorem 3). A sample split supplies nuisances independent of the edge estimates (DRUP-split).
3. *Guarantees and their scope.* With nuisances independent of the log, DRUP is edge-wise doubly robust for propagation over the full-exposure graph (Theorem 1), unbiased on unexposed candidates up to the imputation error at the candidate (Corollary 2), and its expected scores do not respond to interventions on an item’s exposure (Theorem 6). We show by simulation where these guarantees fail.

4. *Evidence.* A semi-synthetic test on KuaiRec with known exposure; a comparison with up to 26 tuned baselines on Coat, Yahoo!R3, KuaiRand and KuaiRec with user-level tests; and real-data thinning audits with a negative control. The properties that follow from linearity alone (attributions, certificates, private release, capped allocation) are in the Supplementary Material.

Outline. Section 3 reviews related work, Section 5 presents the method, Section 7 the datasets, baselines and protocol, Section 8 the results, and Section 9 the theoretical and practical implications and limits.

3. Related work

Graph-based recommendation. Graph neural recommenders propagate embeddings over the user–item graph. LightGCN (He et al., 2020) removes feature transformations and non-linearities, which reduces propagation to a weighted sum of powers of a normalised adjacency; UltraGCN (Mao et al., 2021) approximates infinite-layer propagation, and SGL (Wu et al., 2021) and SimGCL (Yu et al., 2022) add contrastive views. Closed-form models are strong competitors: EASE (Steck, 2019) and EDLAE (Steck, 2020) learn an item–item matrix with a constrained diagonal, which, like our diagonal correction, removes an item’s own contribution to its score; GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023) cast collaborative filtering as graph signal filtering. Carefully tuned simple baselines often match graph models (Anelli et al., 2023). All of these propagate over the *logged* graph. DRUP keeps the linear propagation form but propagates over an unbiased estimate of the full-exposure graph.

Debiasing in recommendation. Logs are missing not at random (Marlin and Zemel, 2009); surveys organise the resulting biases and cast debiasing as causal inference (Chen et al., 2023; Gao et al., 2024). Propensity-based methods re-weight the loss (Schnabel et al., 2016; Swaminathan and Joachims, 2015; Saito et al., 2020). Doubly robust learning adds an imputation model (Wang et al., 2019), and later variants reduce its bias and variance (Guo et al., 2021; Dai et al., 2022; Li et al., 2023). A small uniform sample can supervise the debiasing (Chen et al., 2021). Causal approaches to popularity bias use counterfactual inference (Wei et al., 2021), confounder adjustment (Zhang et al., 2021) or disentangled representations (Zheng et al., 2021); recent work in this journal studies popularity debiasing from both sides (Boratto et al., 2021), with a controllable trade-off (Okamura et al., 2024), through invariant learning (Bai et al., 2025) and through causal deconfounding of multi-criteria ratings (Guo et al., 2026). These methods correct the loss or the output of a model; DRUP applies inverse-propensity and DR estimation to the *edges* of the propagated graph.

Debiasing inside graph propagation. NAVIP (Kim et al., 2022) weights each edge by its inverse propensity inside LightGCN. r-AdjNorm (Zhao et al., 2022) tunes the adjacency normalisation that controls the amplification of popularity, and graph convolutions provably amplify popularity bias (Chen et al., 2024). APDA (Zhou et al., 2023), CAGED (Que et al., 2025) and DPAA (Islam et al., 2026) learn aggregation weights. These methods change *which* adjacency is propagated. None analyses what happens to an unbiased edge estimate when the re-weighted adjacency is multiplied with itself over several hops. Theorem 5 shows that the product is then biased by a term

of order $1/p$ that concentrates on rarely exposed items. Our linear baseline over the logged graph searches the normalisation exponent and is therefore a training-free r-AdjNorm; we also train r-AdjNorm and NAVIP.

Unbiased estimation of polynomials under missingness. An unbiased estimator of a polynomial in unknown quantities must avoid products of the same noisy measurement, as in U-statistics (Hoeffding, 1948). For Bernoulli-masked data the masked Gram matrix needs a $1/p^2$ correction off the diagonal and $1/p$ on it (Lounici, 2014); our item operator is the analogue with heterogeneous propensities. Möbius inversion on the partition lattice (Rota, 1964) turns homomorphism counts into injective counts and is the combinatorial counterpart of our K -hop construction. Non-backtracking operators (Krzakala et al., 2013) and self-avoiding walks (Hopkins and Steurer, 2017) discard repeated walks; our correction keeps them and replaces each repeated factor by its unbiased counterpart. In recommendation, Schnabel and Bennett (2020) debias item co-occurrence with propensities. The reduction itself is therefore not new; what is new is its combination with inverse-propensity and DR edge estimates and degree weights, the exact bias of the uncorrected operators, and the consequences for exposure invariance. Our treatment of nuisances follows double machine learning (Chernozhukov et al., 2018).

Fairness, explanation, robustness and privacy. Fairness notions cover users and items, groups and individuals, and outcomes and exposure (Wang et al., 2023; Amigó et al., 2023); item-side exposure can be allocated in proportion to merit (Singh and Joachims, 2018; Diaz et al., 2020), and multi-stakeholder views weigh users, providers and the platform (Abdollahpouri et al., 2020; Ekstrand et al., 2022). DRUP contributes a causal property of the merit

estimate, exposure invariance, which is not a fairness criterion. Graph explainers search for important or minimal edge sets (Ying et al., 2019; Lucic et al., 2022; Ghazimatin et al., 2020), recommenders can be attacked by injected profiles (Lam and Riedl, 2004; Li et al., 2016; Fang et al., 2018), and private recommenders perturb statistics or matrix completion (McSherry and Mironov, 2009; Jain et al., 2018). Any linear operator with fixed nuisances admits exact removal attributions, closed-form certificates and a jointly private release of its item operator (Balle and Wang, 2018; Hsu et al., 2014); these are in the Supplementary Material.

4. Problem setting

Users $u \in [m]$, items $i \in [n]$ and pairs $e = (u, i)$ carry a binary potential outcome $Y_e \in \{0, 1\}$ (whether u would like i if it were shown) and an observation indicator $O_e \in \{0, 1\}$. The log reveals O and $O \odot Y$. What O records depends on the platform. On KuaiRec it is an impression chosen by the system. On Coat and Yahoo! R3 it is a rating that the user chose to give, so the “exposure” mechanism modelled by p is then user self-selection rather than the platform’s display policy. All statements below are relative to the mechanism that generates O . We keep the word exposure for brevity.

Assumption 1 (Conditionally independent exposure). *Conditionally on Y , the exposures O_e are independent Bernoulli variables with propensities $p_e = \Pr(O_e = 1 \mid Y) > 0$.*

The propensity may depend on the pair’s own outcome, so the log may be missing not at random. All expectations below are over O given Y .

Assumption 2 (Fixed nuisances). *The propensity estimates $\hat{p}$, the imputation $\hat{Y} \in [0, 1]^{m \times n}$, the edge weights C and the constants a, b of the score below are fixed given Y , i.e., they do not depend on O (for instance, they are fitted on an independent log). We use $C_{ui} = d_u^{-\alpha} d_i^{-(1-\alpha)}$ with degrees d taken from $\hat{Y}$.*

The weights C also enter the target below. With degrees from $\hat{Y}$, the target therefore depends on the imputation. A normalisation fixed a priori (a design choice, or degrees from an independent source) makes the target independent of $\hat{Y}$; Section 8.2 compares both.

Remark 1 (Scope of the guarantees and the experimental protocol). All guarantees below are stated under Assumption 2. Our main experimental protocol does not satisfy it, in four ways. (i) $\hat{p}$ and $\hat{Y}$ are cross-fitted over ten random folds of pairs: the nuisances used at a pair of fold k are fitted on the other nine folds (Chernozhukov et al., 2018). This makes the nuisance at a pair independent of that pair’s exposure, which is what a one-hop estimate needs. It does not make the nuisance independent of the exposures of the other edges of a walk. A product of several edge estimates can therefore be biased, even though every single edge estimate is unbiased. (ii) The degree weights sum the imputation over all pairs of a user or an item. They therefore depend on the exposures of all these pairs, including the pair at which they are used. (iii) One of the searched degree sources takes degrees from edge estimates. (iv) The constants a, b are computed from the scores (Section 5). In simulation, the resulting dependence bias is of the same order as the repeated-walk bias. Under this protocol with imputation-based degrees, the corrected operator is not less biased than the uncorrected one (Section 8.1).

In a control where only the degree weights are fixed a priori, the correction works again under cross-fitting. So (ii) and (iii), more than (i), account for the failure. The theorems therefore describe DRUP with nuisances from an independent log, or from the sample split introduced below. For the cross-fitted protocol they give no guarantee. Our experiments with it evaluate accuracy and the empirical behaviour of the scores, not unbiasedness.

With $\tilde{Y} = C \odot Y$, the target is propagation over the full-exposure graph,

$$F^* = a\tilde{Y} + bT^*, \quad T^* = \tilde{Y}\tilde{Y}^\top\tilde{Y}, \quad (1)$$

with fixed constants $a, b \geq 0$. F^* is the linear LightGCN score one would compute if every item had been shown to every user. It is a normative choice. It treats potential outcomes as fixed and ignores preferences that exposure itself shapes (Chaney et al., 2018; Perdomo et al., 2020) (Section 9).

5. DRUP

Edge estimates. With clipped propensities $\bar{p} = \max(\hat{p}, \tau)$,

$$W_e^{\text{IPS}} = \frac{O_e Y_e}{\bar{p}_e}, \quad W_e^{\text{DR}} = \hat{Y}_e + \frac{O_e(Y_e - \hat{Y}_e)}{\bar{p}_e}. \quad (2)$$

Both have mean Y_e when $\bar{p} = p$.

Where the plug-in operator fails. The plug-in three-hop operator $(\tilde{W}\tilde{W}^\top\tilde{W})_{ui}$, with $\tilde{W} = C \odot W$, sums the products $\tilde{W}_{uj}\tilde{W}_{vj}\tilde{W}_{vi}$ over all walks $u-j-v-i$. If $v \neq u$ and $j \neq i$, the three edges are distinct and, by independence, the product is unbiased. The other walks repeat an edge (Figure 1). For $v = u$ the edge (u, j) is traversed twice. For

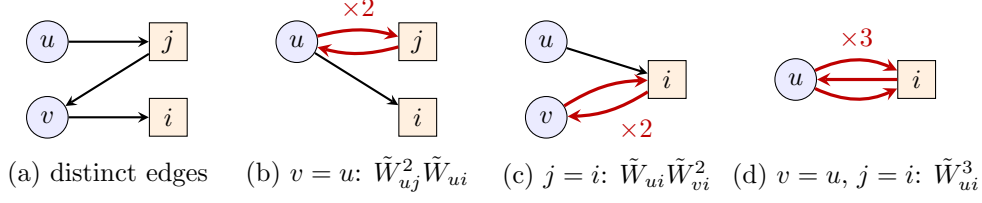


Figure 1: The four kinds of three-hop walks from user u to item i . Only walks of type (a) consist of distinct edges. In (b)–(d) an edge estimate is multiplied with itself (red), and its power has the wrong mean. DRUP replaces each repeated factor $\tilde{W}_e^k$ by $C_e^k W_e$.

$j = i$ the edge (v, i) is traversed twice. For $v = u, j = i$ the edge (u, i) is traversed three times. A repeated factor $\tilde{W}_e^k$ has mean $C_e^k \mathbb{E}[W_e^k] \neq C_e^k Y_e$; for IPS, $\mathbb{E}[W_e^2] = Y_e/p_e$. Because $Y_e^k = Y_e$ for binary outcomes, the unbiased replacement of $\tilde{W}_e^k$ is simply $C_e^k W_e$.

Idempotent walk correction. Let $\Delta = \tilde{W} \odot \tilde{W} - C \odot C \odot W$, with row sums r and column sums κ . The corrected three-hop operator subtracts the excess of each walk type:

$$\hat{T} = \tilde{W} \tilde{W}^\top \tilde{W} - \tilde{W} \odot (r \mathbf{1}^\top - \Delta) - \tilde{W} \odot (\mathbf{1} \kappa^\top - \Delta) - (\tilde{W}^{\odot 3} - C^{\odot 3} \odot W). \quad (3)$$

The second term removes the excess of type (b) walks, the third removes that of type (c), and the last removes that of type (d). The subtracted Δ inside the first two terms avoids counting type (d) twice. The DRUP score is $s = a \tilde{W} + b \hat{T}$. In the experiments $a = 1/c_1$ and $b = \beta/c_3$. Here c_1 and c_3 are the mean absolute values of the one- and three-hop terms over all scored users, and β is selected on validation data. The constants are global, so every user’s scores are the same linear combination of the two terms. Within-user rankings depend on the constants only through b/a , so a data-dependent c_3/c_1 acts as a data-dependent rescaling of the grid of β . However, c_1 and c_3

are computed from the log, so $\mathbb{E}[\hat{T}/c_3] \neq T^*/c_3$ in general, and the score-level statements below hold for fixed a, b . Constants computed from the imputed graph $C \odot \hat{Y}$ satisfy Assumption 2 whenever $\hat{Y}$ does. Section 8.1 compares both choices.

Sample-split variant. A log rarely comes with an independent log for its nuisances. A sample split provides one, at a cost. Draw a random set A of pairs, each pair with probability q . Fit $\hat{Y}$ (and $\hat{p}$, where it is estimated) on the pairs in A only. Replace the edge estimates of the pairs in A by $\hat{Y}$, take the degrees from $\hat{Y}$, and compute c_1 and c_3 on the imputed graph $C \odot \hat{Y}$. Conditionally on A and on the exposures in A , the remaining edge estimates are independent of every nuisance. Assumption 2 then holds and the results below apply, with $\xi_e = \hat{Y}_e - Y_e$ on the pairs in A . The price is a first-order imputation bias on a fraction q of the pairs, and fewer exposures for the edge estimates. We call this variant DRUP-split and use $q = 0.2$. The guarantee holds for every fixed configuration. It does not cover the selection of hyperparameters on validation data, which every method in our comparison does and which is outside the guarantee of any estimator.

Public-operator form. Define the diagonal-corrected item Gram matrix $G = \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$. Then

$$\hat{T}_u = \tilde{w}_u G - \tilde{w}_u \odot (r_u \mathbf{1}^\top - 2\zeta_u) - (\tilde{w}_u^{\odot 3} - c_u^{\odot 3} \odot w_u), \quad (4)$$

where $\zeta_u = \tilde{w}_u^{\odot 2} - c_u^{\odot 2} \odot w_u$ and lower-case rows belong to user u . Equation (4) matches (3) to machine precision. G is the only object shared across users (Algorithm 1). Its off-diagonal entries are products of distinct edges; its di-

Algorithm 1 DRUP (training-free)

Require: $\log(O, O \odot Y)$; hyper-parameters $\tau, \alpha, \lambda, \beta$

- 1: fit $\hat{p}$ (logistic exposure model) and $\hat{Y}$ (IPS-weighted additive ridge, optionally plus an IPS-weighted low-rank ALS residual), cross-fitted over ten folds of pairs
 - 2: $\bar{p} \leftarrow \max(\hat{p}, \tau)$; $W \leftarrow \hat{Y} + O \odot (Y - \hat{Y})/\bar{p}$
 - 3: $C \leftarrow d_u^{-\alpha} d_i^{-(1-\alpha)}$ with d from $\hat{Y}$
 - 4: $G \leftarrow \tilde{W}^\top \tilde{W} - \text{diag}(\tilde{W}^\top \tilde{W}) + \text{diag}(\mathbf{1}^\top (C \odot C \odot W))$ $\triangleright$ public operator
 - 5: **for** each user u (locally) **do**
 - 6: $\hat{T}_u \leftarrow \text{Eq. (4)}$; $s_u \leftarrow a \tilde{w}_u + b \hat{T}_u$; rank unexposed items by s_u
 - 7: **end for**
-

agonal is corrected from $\sum_u \tilde{W}_{ui}^2$ to $\sum_u C_{ui}^2 W_{ui}$, the analogue of the diagonal correction of a Gram matrix estimated from masked data (Lounici, 2014).

Control-variate family. More generally, $W_e^\lambda = O_e Y_e / \bar{p}_e + \lambda(\hat{Y}_e - O_e \hat{Y}_e / \bar{p}_e)$ is unbiased for every $\lambda \in [0, 1]$ under correct propensities. It interpolates between IPS ($\lambda = 0$) and DR ($\lambda = 1$, the only member that is also robust to propensity errors). It equals the DR estimate with the shrunk imputation $\lambda \hat{Y}$. With $\bar{p} = p$ its variance is $\frac{1-p}{p}(Y - \lambda \hat{Y})^2$. All results below hold for every member, and λ is selected on validation data.

Imputation and cost. Theorem 1 holds for any fixed $\hat{Y}$. We use an additive model $\mu + b_u + b_i$ fitted by IPS-weighted ridge regression, or that model plus a rank-32 residual fitted by IPS-weighted alternating least squares. At three hops the correction costs one $n \times n$ Gram product and $O(mn)$ reductions, which is the cost of the uncorrected operator. The five-hop correction is 29

times slower (Section 8.1). The dense W and G need $O(mn + n^2)$ memory. This limits the present implementation to catalogues of about 10^4 – 10^5 items.

6. Theory

Proofs are in Supplementary Section S1. Let $\varepsilon_e = |Y_e - \hat{Y}_e|$.

6.1. Unbiasedness and its limits

Theorem 1 (Edge-wise double robustness). *Under Assumptions 1–2, let $\xi_e = (p_e/\bar{p}_e - 1)(Y_e - \hat{Y}_e)$. Then $\mathbb{E}[\hat{T}_{ui}]$ equals T_{ui}^* with every distinct edge contributing $Y_e + \xi_e$ once. In particular, if every edge has a correct propensity or a correct imputation, then $\mathbb{E}[s] = F^*$.*

Recommendations rank the *unexposed* candidates of a user, and there the candidate’s own edge estimate is deterministic ($W_{ui} = \hat{Y}_{ui}$ for DR, 0 for IPS). The following corollary states what DRUP estimates on the candidate set.

Corollary 2 (Unexposed candidates). *Under the conditions of Theorem 1 with $\xi \equiv 0$,*

$$\begin{aligned} \mathbb{E}[s_{ui} \mid O_{ui} = 0] &= F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) g_{ui}, \\ g_{ui} = \frac{\partial F_{ui}^*}{\partial Y_{ui}} &= C_{ui} \left(a + b \left(\sum_{v \neq u} C_{vi}^2 Y_{vi} + \sum_{j \neq i} C_{uj}^2 Y_{uj} + C_{ui}^2 \right) \right) \geq 0. \end{aligned}$$

For IPS edges the same holds with $\hat{Y}_{ui} = 0$. The residual is the imputation error at the candidate itself. If p_e depends on Y only through Y_e , the log carries no information on Y_{ui} when $O_{ui} = 0$, so the residual is not identified and no estimator can remove it.

At three hops the walks through (u, i) are exactly those counted in g_{ui} . With IPS edges they vanish on unexposed candidates, so the uncorrected IPS operator has no repeated-walk bias there. From five hops on, walks repeat edges that are not incident to the candidate (for example $u-j-v-j-v-i$), and the bias reappears (Table 3).

Theorem 3 (Any odd number of hops). *For $K = 2r + 1$, a walk $U_0 - J_1 - U_1 - \dots - U_r - J_{r+1}$ has user variables $U_0, \dots, U_r$ and item variables $J_1, \dots, J_{r+1}$, with $U_0 = u$ and $J_{r+1} = i$. The equality pattern of an assignment is a pair π of set partitions of the user and item variables, and π fixes the multiplicity k_e of every distinct edge. Let $g_\pi(\sigma)$ be the unconstrained contraction, over the blocks of a coarser pattern $\sigma \geq \pi$, of $\prod_e C_e^{k_e} W_e$ (corrected) or of $\prod_e \tilde{W}_e^{k_e}$ (plug-in), the product running over the distinct edges of π . Then*

$$\hat{T}^{(K)} = \tilde{W}(\tilde{W}^\top \tilde{W})^r + \sum_{\pi: \exists k_e > 1} \sum_{\sigma \geq \pi} \mu(\pi, \sigma) (g_\pi^{\text{corr}}(\sigma) - g_\pi^{\text{plug}}(\sigma)),$$

with μ the Möbius function of the product of the two partition lattices, satisfies $\mathbb{E}\hat{T}^{(K)} = (\tilde{Y}\tilde{Y}^\top)^r \tilde{Y}$ under the conditions of Theorem 1 with $\xi \equiv 0$.

The number of pairs (π, σ) is 5, 99 and 2,790 for $K = 3, 5, 7$, and for $K = 3$ the construction reduces to Eq. (3). Theorem 1, Corollary 2 and Theorem 6 carry over to any odd K .

Assumption 1 excludes slates of fixed size, where exposures in a user’s history are negatively dependent (Coat, for instance, collects exactly 24 ratings per user). The next result bounds the resulting bias.

Proposition 4 (Exposure dependence within users). *Suppose that exposures are independent across users but may be dependent within a user’s row, with*

$|\text{Cov}(O_e, O_f \mid Y)| \leq \rho p_e p_f$ for distinct e, f in the same row. With correct propensities,

$$|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq \rho \sum_{j \neq i} C_{uj} \left(\sum_{v \neq u} C_{vj} C_{vi} \varepsilon_{vj} \varepsilon_{vi} + C_{uj} C_{ui} \varepsilon_{uj} \varepsilon_{ui} \right).$$

For slates of k items drawn uniformly without replacement, $\rho \leq 1/k$, and $\rho \leq 1$ for any negatively dependent design. For DR edges the bias is of second order in the imputation error; for IPS edges ($\varepsilon = Y$) it is of first order.

Theorem 5 (Repeated-walk bias of uncorrected propagation). *Let the propensities be correct and let (u, i) be a candidate with $O_{ui} = 0$. Relative to the conditional target of Corollary 2, the uncorrected DR operator $\tilde{W}\tilde{W}^\top\tilde{W}$ has conditional bias*

$$C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i) - C_{ui}^3\hat{Y}_{ui}(1 - \hat{Y}_{ui}^2),$$

$$K_i = \sum_{v \neq u} C_{vi}^2 V_{vi}, \quad R_u^{(i)} = \sum_{j \neq i} C_{uj}^2 V_{uj},$$

where $V_e = \text{Var}(W_e) = \frac{1-p_e}{p_e} \varepsilon_e^2$. Moreover: (a) $K_i \geq (1/\max_v p_{vi} - 1) \sum_{v \neq u} C_{vi}^2 \varepsilon_{vi}^2$, which is unbounded as the item's propensities vanish; (b) with clipped propensities $\bar{p} = \max(p, \tau)$, $\tau \leq 1/2$, the excess $C_e^2(\mathbb{E}[W_e^2] - \mathbb{E}[W_e])$ of a doubly traversed edge is at most $C_e^2(1/\tau - 1)\varepsilon_e^2$, with equality at $p_e = \tau$; the worst-case repeated-walk bias is therefore $\Theta(1/\tau)$; (c) for IPS edges the conditional bias vanishes at three hops. DRUP has zero bias relative to the same target.

The within-user ranking is affected through $C_{ui}\hat{Y}_{ui}K_i$, which up-weights items whose other raters were rarely exposed. The term $R_u^{(i)}$ depends on i only through one term.

Variance and clipping. The variance of an edge estimate scales with the imputation error ε^2 , not with Y^2 as for IPS. The clip τ trades variance against a clipping bias, which vanishes when $\hat{Y}$ is accurate. The bounds (Theorem S1 in Supplementary Section S1) are worst-case and loose by a factor of 70 to 8,000 in simulation.

Definition 1 (Exposure elasticity). For an intervention $do(p_i \leftarrow \pi p_i)$ on item i 's exposure mechanism, after which the estimator uses the post-intervention propensities ($\hat{p}_i \leftarrow \pi \hat{p}_i$) and otherwise the same nuisances, $\eta_i = \partial \log \mathbb{E}[s_{ui}] / \partial \log \pi$.

Theorem 6 (Exposure invariance). *Under Assumptions 1–2 and $\pi p \geq \tau$, DRUP has $\eta_i = 0$, also conditionally on $O_{ui} = 0$: the expected score of every candidate is unchanged when item i 's exposure mechanism is scaled. Logged-graph propagation with fixed weights C has $\eta_i \geq 1$ (popularity amplification); normalising by logged degrees partly offsets it. Uncorrected DR propagation has $\eta_i < 0$ whenever $\hat{Y}_{ui} K_i > 0$ (over-correction).*

Exposure invariance is a statement about expected scores. It is not a distributive fairness criterion. It is also conditional: when many propensities lie below the clip, $\pi p \geq \tau$ fails and invariance is lost (Section 8.4).

6.2. Structural properties

Given its nuisances, s is a fixed linear function of the edge estimates. Section S3 of the Supplementary Material states and proves the consequences. They hold for every linear propagation operator, corrected or not: exact attributions and minimal counterfactual explanations with frozen nuisances

(Proposition S2, Theorem S3), conservative certificates against injected profiles (Theorem S4), joint differential privacy when the population-level nuisances, the hyper-parameters and the score constants are public (Theorem S5), and an exposure-capped allocation solved exactly by min-cost flow (Theorem S6). Because they are not specific to DRUP, we report their empirical behaviour in the Supplementary Material (Section S4) and use only the allocation in the main text, as a decision that compares scores across users (Section 8.2).

7. Experimental design

7.1. Datasets

Table 2 describes the four benchmarks. Each pairs a biased training log with an unbiased test set, which is what identifies the effect of a debiasing method: a random sample of ratings (Coat, Yahoo!R3), random impressions (KuaiRand) or a fully observed matrix (KuaiRec). All four are public.

Coat (Schnabel et al., 2016) has 290 users, 300 items and 6,960 self-selected ratings (24 per user, 8.0% dense). The test data are 16 ratings per user on uniformly random items; we rank the items the user did not rate (14.7 candidates on average), and 225 users have a positive candidate. The dataset ships propensities, which we use. *Yahoo!R3* (Marlin and Zemel, 2009) has 311,704 self-selected ratings of 15,400 users on 1,000 songs (2.0% dense) and 54,000 ratings of ten random songs for 5,400 of the users, in the version distributed with (Chen et al., 2021). A random 5% of the test users is held out and used only to fit an outcome-dependent propensity model (Schnabel et al., 2016); it did not improve the propagation operators on validation data, so

all reported results use a logistic exposure model. The remaining 5,130 users have ten candidates each, and 2,315 of them a positive. *KuaiRand-Pure* (Gao et al., 2022b) is the only dataset that records platform exposure rather than self-selection: O is an impression of the standard feed and Y a click. We keep a random 10,000 of the users who also received randomly exposed videos, and all 7,583 videos; the log has 511,863 impressions (0.68% dense, click rate 47%). The random impressions of the same period, restricted to videos the user was not shown in the standard log, are the test set (43.6 candidates per user, 17.5% clicked; 8,347 users have a positive). *KuaiRec* (Gao et al., 2022a) provides the big matrix (7,176 users, 10,728 items, 10.3M views, 13.4% dense) as the log and a fully observed small matrix (1,411 users, 3,327 items) as the test set; we rank the 3,314 small-matrix items per user that the user never viewed in the log (154 positive on average).

Positives are ratings ≥ 4 (Coat, Yahoo! R3), a watch ratio ≥ 2 (KuaiRec) and a click (KuaiRand). A random ranking reaches nDCG@5 of 0.306 on Coat and 0.352 on Yahoo! R3, nDCG@10 of 0.328 on KuaiRand and nDCG@20 of 0.046 on KuaiRec, so the candidate sets of Coat and Yahoo! R3 compress metric differences. On Coat and Yahoo! R3 “exposure” means that the user chose to rate the item, and exposure invariance is invariance to that choice, not to a platform’s allocation.

7.2. Baselines

We compare DRUP with up to 26 baselines, all run by us on the same datasets, splits and validation data; no number is taken from the literature. *Non-personalised*: popularity and the imputation $\hat{Y}$ alone. *Trained, pointwise loss*: MF, IPS-MF (Schnabel et al., 2016), DR-MF, DR-JL (Wang

et al., 2019), MRDR (Guo et al., 2021), iALS (Hu et al., 2008), LightGCN and DR-LightGCN. *Trained, pairwise or contrastive loss*: MF, PDA (Zhang et al., 2021), MACR (Wei et al., 2021), LightGCN (He et al., 2020), r-AdjNorm (Zhao et al., 2022), NAVIP (Kim et al., 2022) and SimGCL (Yu et al., 2022). *Training-free, logged graph*: linear LightGCN (with the searched normalisation exponent it is a linear r-AdjNorm), EASE (Steck, 2019), GF-CF (Shen et al., 2021) and BSPM (Choi et al., 2023). *Training-free, debiased graph*: IPS adjacency (NAVIP-style), IPS with the walk correction, EASE, GF-CF and BSPM on the DR graph, uncorrected DR adjacency and the five-hop versions. DRUP-split (Section 5) is run over five draws of the split-off pairs. Grids and training details are in Supplementary Section S2.

How recent are the baselines? The most recent are BSPM (2023), SimGCL, r-AdjNorm and NAVIP (2022) and MRDR (2021), which are among the strongest published graph, contrastive and debiased models for implicit feedback and which are open for re-implementation on the biased-log-plus-random-test protocol; DR-JL (2019) is included because the DR baseline is the direct ancestor of our edge estimate. We did not run StableDR, DICE, CausE or AutoDebias because they either need the uniform data for training (CausE, AutoDebias), which our protocol keeps for selection and testing, or are covered by the corresponding DR and popularity baselines (StableDR, DICE).

Large language models. We did not compare with LLM-based recommenders. Our pipeline uses only interaction data: the four benchmarks come with no item or user text that we use, and the question studied here is how to estimate a propagation operator under exposure bias, for which the relevant

comparators are graph operators. A comparison with LLM-based recommenders would need a dataset that has item text and a random-exposure test set, which none of our benchmarks provides together, and it would test a different hypothesis, that text helps. We list this as a limitation (Section 9).

7.3. Protocol

The unbiased data are split into validation (30% of each user’s candidates on Coat, 30% of the test users otherwise) and test parts over 10 (Coat) or 5 random splits. Every method selects its configuration on the validation part of each split, and trained models also select their epoch there: each split stops on its own validation score and never sees the validation or test users of another split. Nuisances are cross-fitted over ten folds of pairs, except for the sample-split variants, whose imputation (and propensity model, where it is estimated) is fitted on a random 20% of the pairs; only the latter satisfy the assumptions of the theory (Remark 1). We report nDCG@5 on Coat and Yahoo!R3, nDCG@10 on KuaiRand and nDCG@20 on KuaiRec (AUC and Recall are in the Supplementary Material) as the mean and standard deviation over splits. Differences are tested at the level of users, with a Holm correction over all comparisons of a dataset; paired tests over overlapping splits are anti-conservative (Nadeau and Bengio, 2003), so we add the corrected resampled test over splits and equivalence tests for the key comparisons (Supplementary Table S19). The tests treat the training log, the fitted nuisances and the trained models as fixed. Because progress claims in top- N recommendation have repeatedly failed against tuned simple baselines (Ferrari Dacrema et al., 2019; Rendle et al., 2019; Anelli et al., 2023), every method is tuned on the same validation data, and the number of con-

figurations each searched is reported (Supplementary Section S2).

8. Results

8.1. Simulation: does the correction work, and under which conditions?

This section addresses RQ1 and RQ2. Table 3 uses a synthetic MNAR model with known propensities: 30 users, 40 items, low-rank preferences, and exposures driven by user activity, item popularity and the outcome itself (median propensity 0.06). We draw 20,000 exposure matrices for three hops and 4,000 for five hops.

The bias is large and the correction removes it (RQ1). Over all entries, the uncorrected IPS and DR operators are biased by 2.3 and 1.8 times the mean target at three hops and by 15 and 8 times at five hops. With the correction the residual bias (0.02–0.06) is at the Monte-Carlo error. On the unexposed candidates that are actually ranked, Corollary 2 predicts that IPS is unbiased at three hops but not at five, and the uncorrected DR operator is biased at both depths; the simulation agrees (IPS 0.029 at three hops and 4.7 at five; DR 1.1 and 5.1). The exposure elasticity is +1.78 for the logged graph, −0.19 for DR adjacency and +0.005 for DRUP. The K -hop engine matches the closed-form three-hop operator to 7×10^{-15} and brute-force enumeration of five-hop walks to 10^{-13} . On a dense matrix of KuaiRec’s size (7,176 users, 10,728 items, 1,411 scored users, four CPU cores) the corrected three-hop operator takes 1.9s against 1.4s for the plug-in one, and the corrected five-hop operator 85s against 2.9s: the correction is exact but not free beyond three hops.

Unbiasedness does not buy ranking accuracy. The rank agreement of one realisation with the target is higher for the uncorrected DR operator (Kendall $\tau = 0.20$) than for DRUP (0.12). On unexposed candidates the bias is $C_{ui}\hat{Y}_{ui}(R_u^{(i)} + K_i)$ (Theorem 5), which adds a positive multiple of the imputation to the score, and here the imputation is informative. For IPS edges, whose bias does not involve the imputation, the correction leaves the rank agreement unchanged.

The guarantees do not survive data-dependent weights (RQ2). Table 4 measures the guarantees under the ways nuisances are fitted in practice. With nuisances from an independent log the corrected operator is at the Monte-Carlo error (0.01–0.10), while the uncorrected operators are biased by 0.09–2.2 times the mean target. Under the cross-fitted protocol of our main experiments the advantage disappears: the corrected operator is biased by 0.17–0.80 and the uncorrected one by 0.11–0.61, because the degree weights depend on the exposures of the walk edges. A control that fixes the degree weights a priori (Table S18 in the Supplementary Material) isolates the cause: with fixed weights, cross-fitting alone preserves the correction (IPS with the correction 0.05 against 1.10 uncorrected; DRUP 0.05 against 0.89), whereas degrees from the cross-fitted $\hat{Y}$ or from the edge estimates destroy it (0.64–0.80 and 0.74–0.82). Fitting the nuisances in sample breaks the correction even with fixed weights for DR edges (0.23). The sample split restores the guarantee: relative to the target of Theorem 1, the bias of DRUP-split is 0.007–0.065, and relative to the full-exposure target 0.07–0.18 at $q = 0.2$, against 0.07–1.09 for the uncorrected operator under the same split. The data-dependent constants matter too: with independent nuisances and DR

edges, the full score is biased by 0.18–0.35 when c_1, c_3 are computed from the scores and by 0.02–0.11 when they come from the imputed graph. Stress tests (fixed-size slates, misspecified propensities, bounds on the variance) are in the Supplementary Material.

8.2. Score values on a semi-synthetic KuaiRec log (RQ3, part 1)

On real benchmarks the full-exposure target F^* is unknown, so they cannot show whether score values are right. We therefore use KuaiRec’s fully observed small matrix (1,411 users, 3,327 items, 4.6% positives) as the potential outcomes Y and draw exposure logs from a known MNAR mechanism fitted to the big matrix, $\text{logit } p_{ui} = c + z_u + 1.5z_i + Y_{ui}$, with z the standardised logit exposure rates of the user and the item in the big log and c set for a density of 13% as in the big matrix (median propensity 0.064, all propensities at least $\tau = 0.02$). The target uses weights C fixed a priori (degrees of Y , $\alpha = 0.5$) and a, b fixed from the target. Each estimator uses either the same fixed C or degrees from its imputation (as in our real-data configurations), a rank-16 IPS-weighted imputation, and nuisances from an independent log, cross-fitted on the same log, or from the 20% sample split. Table 5 reports score errors and three uses of the scores.

Score values. With true propensities and fixed weights, the correction lowers the relative RMSE of the three-hop term by a factor of 4 to 12 (IPS 22.8 to 2.9, DR 28.2 to 6.3 with an independent log; DR 35.3 to 8.5 cross-fitted): the repeated factors of rarely exposed edges dominate the error as well as the bias. With 30 draws the remaining three-hop bias of the corrected operators is within Monte-Carlo error; at five hops the relative bias drops from 3.46

(DR) to 0.32 (DRUP). The difference shows in a quantity that uses score values: the sum of the scores of a fixed allocation estimates its full-exposure utility, and the uncorrected operators overestimate it by 18–29% (random allocation) and 10–19% (popular items), whereas IPS with the correction and DRUP are within 2% of the truth. This holds with an independent log and, because C is fixed, also with cross-fitted nuisances (DRUP $-0.1 \pm 1.6\%$ against DR $+28.6 \pm 4.7\%$). The sample split removes the dependence but imputes a fifth of the pairs, and its imputation bias shifts the corrected estimates up by 38–47% (uncorrected: 53–64%).

What does not change. Two uses of the scores are unaffected. The exposure-capped allocation, which compares scores across users, loses 79–84% of the achievable full-exposure utility under every estimator: per-pair noise, not the repeated-walk bias, decides which pairs are allocated. Rankings do not improve either: with imputation-based weights, DR adjacency and DRUP reach nDCG@20 of 0.377 and 0.370. With weights taken from the imputation, finally, every operator estimates a different, imputation-indexed quantity; with an independent log or cross-fitted nuisances it is off by 40–57% in utility against the fixed-weight target, corrected or not, so score values need weights fixed a priori. With estimated propensities all estimators overstate utility by 100–540%, and the propensity model, not the walk correction, dominates the error (Table S14).

A sparser log. At a density of 3% (median propensity 0.008, 58% of the propensities below the clip $\tau = 0.01$) the conditions of the theorems fail, and clipping dominates (Table S15). The correction still lowers the three-hop RMSE by a factor of 7 to 32, but every utility estimate is biased: IPS with

the correction by -28% , DR adjacency by $+331\%$ and DRUP by $+164\%$ with an independent log, in line with the clipping bias of Theorem S1(c).

8.3. Accuracy on unbiased test data (RQ3, part 2)

Tables 6 and 7 report accuracy and the user-level tests; AUC and Recall are in the Supplementary Material and lead to the same conclusions.

The walk correction does not change accuracy. DRUP and the uncorrected DR adjacency with the same grid are not significantly different (Coat $+0.0014$, KuaiRec -0.0001 , $p_{\text{Holm}} = 1$). Only KuaiRec and, marginally, Coat can test equivalence: within ± 0.005 nDCG the two operators are equivalent on KuaiRec ($p_{\text{TOST}} < 10^{-3}$, also at ± 0.002) and marginally so on Coat ($p_{\text{TOST}} = 0.034$, but 0.38 at ± 0.002 ; Table S19). The margin is a convention, about twice the split-to-split standard deviation of the methods on KuaiRec (0.002–0.004). On Yahoo!R3 and KuaiRand the two operators coincide exactly, so there is nothing to test: validation selects the IPS end of the control-variate family ($\lambda = 0$) on every split, and IPS edges have no three-hop repeated-walk bias on unexposed candidates (Corollary 2). At five hops, where the correction is active for both edge types, the corrected operator is lower by 0.0003 nDCG@20 on KuaiRec ($p_{\text{Holm}} = 0.17$) and by 0.008 nDCG@5 on Coat ($p_{\text{Holm}} = 1$); five hops do not add accuracy over three.

The sample split costs accuracy on sparse logs. DRUP-split, the only variant covered by the guarantees, reaches 0.632 ± 0.004 on KuaiRec over five draws of the split-off pairs (0.006 below cross-fitted DRUP), 0.544 ± 0.011 on Coat (0.527 to 0.558: which fifth of 24 ratings per user feeds the nuisances matters),

0.440 ± 0.001 on KuaiRand (0.009 below) and 0.595 ± 0.004 on Yahoo!R3 (0.062 below, $p_{\text{Holm}} < 10^{-3}$), where a fifth of the few logged exposures goes to the nuisances and the imputation that replaces them is poor. DRUP-split and DR adjacency under the same split are not significantly different on any dataset.

Which method is most accurate depends on the dataset. KuaiRec. Training-free propagation over the DR graph reaches nDCG@20 of 0.638, ahead of every trained model at the user level but by little over the best: DR-JL and MRDR (0.633; $+0.004$, $p_{\text{Holm}} = 0.003$). With the Nadeau-Bengio test over splits, adjusted over the same family, neither margin is significant ($p_{\text{NB,Holm}} = 1.0$ and 0.34), and both are smaller than the seed spread of these models (0.009 and 0.015). The margin over pointwise LightGCN (0.627; $+0.010$) is significant under both tests. GF-CF on the DR graph performs as well as DRUP (0.637). Propagating the imputation alone, with no observed residual, reaches 0.631: two thirds of the 0.018 that DRUP gains over the imputation (0.619) comes from propagating the imputation. *KuaiRand.* DRUP, DR adjacency and IPS adjacency coincide (nDCG@10 0.449), are not significantly different from GF-CF (0.450), BSPM (0.450) or iALS (0.453), and are significantly behind the BPR graph models: r-AdjNorm 0.459, LightGCN 0.457, NAVIP 0.457 ($p_{\text{Holm}} < 10^{-3}$). *Yahoo!R3.* The imputation alone is below popularity (0.516 against 0.533), so validation selects IPS edges ($\lambda = 0$) with cross-fitted W -degrees, outside Assumption 2; DRUP (0.657) is on par with the logged graph (0.659) and iALS (0.663) and significantly behind r-AdjNorm (0.674; -0.019 , $p_{\text{Holm}} < 10^{-3}$), LightGCN and NAVIP. *Coat.* iALS is the most accurate (0.581), then BSPM, DR-LightGCN and

the BPR graph models (0.566–0.573); DRUP reaches 0.547. After the Holm adjustment over 124 comparisons DRUP is ahead of pointwise MF, IPS-MF, DR-JL and MRDR only.

Tuning budgets and seed noise. The training-free operators search 384 to 6,300 configurations against 6 to 108 for the trained models (Supplementary Table S1). At the six configurations of DR-JL, the expected KuaiRec score of DRUP is 0.618 against 0.633 (Supplementary Table S2); counted in validation evaluations, which include every epoch of a trained model, DRUP with the 103 evaluations of DR-JL reaches 0.636. Re-training the split-0 selection of the trained models with other seeds changes their score by up to 0.013 on Yahoo! R3 and 0.05 on Coat (r-AdjNorm 0.521–0.566), and by 0.009 and 0.015 on KuaiRec for DR-JL and MRDR, so differences on Coat and the KuaiRec margin over DR-JL and MRDR should be read with this in mind. The loss matters more than the architecture, in opposite directions: pairwise training is decisive on Yahoo! R3 and KuaiRand (pointwise MF variants stay near 0.51 and 0.37), while on KuaiRec the pointwise models beat their BPR versions. No single trained baseline is strong on all datasets.

8.4. *Exposure invariance on real data (RQ4)*

Table 8 tests Theorem 6 with a known intervention on exposure. The operators use the cross-fitted protocol, which the theorem does not cover (Remark 1), so this is an empirical check. The logged graph behaves as predicted for correlational propagation: with frozen nuisances $\eta = +1.00$ to $+1.03$ on all four datasets, and thinned items drop by 0.07–0.24 in mean within-user rank. On KuaiRec and Coat, DRUP is invariant to within a

few hundredths ($\eta = -0.012$ and -0.008), also when its nuisances are re-fitted on the thinned log ($|\eta| \leq 0.022$). IPS adjacency is far from invariant ($\eta = +0.14$ on KuaiRec, $+0.90$ on Coat). The uncorrected DR adjacency is as invariant as DRUP on these data. On Yahoo! R3 and KuaiRand the selected configurations are not invariant ($\eta = +0.96$ and $+0.64$): validation selects IPS edges ($\lambda = 0$) with a clip that binds for 82% of the logged pairs on Yahoo! R3 and for 57% of the treated pairs on KuaiRand. A clipped IPS edge is $O_e Y_e / \tau$, proportional to the exposure itself, so DRUP inherits the elasticity of the logged graph. On Yahoo! R3, lowering the clip to $5 \cdot 10^{-4}$ reduces η to $+0.18 \pm 0.11$ at a cost of 0.044 nDCG@5 (Supplementary Material).

How informative is the audit? With frozen nuisances the audit updates the propensities to the known $p/2$, and an unbiased edge estimate is then unaffected by construction; a value near zero shows that the estimator is used as intended, not that the scores are right. Table 9 adds controls. Popularity re-fitted on the thinned log has $\eta \approx 1$ and the imputation alone $\eta \approx 0$ on Coat and Yahoo! R3 (-0.16 on KuaiRec), which brackets the scale. The negative control, which keeps the propensities of the original log, should push a debiasing estimator towards $\eta = 1$. It does for IPS adjacency on all four datasets and for every operator on Yahoo! R3 and KuaiRand, whose selected configurations use IPS edges, but not for DR adjacency or DRUP on Coat ($\eta = -0.008$) or KuaiRec (-0.088). On Coat the scores are almost a function of the imputation (rank correlation 0.98–0.99 with $\hat{Y}$), which is fitted on exposed pairs and reacts little to a halved exposure of some items, so a near-zero η there says little beyond the invariance of $\hat{Y}$. DRUP-split, whose edge estimates are less dominated by the imputation (rank correlation 0.76

on Coat, 0.58 on Yahoo!R3, 0.55 on KuaiRand), is invariant on Coat and KuaiRec ($|\eta| \leq 0.022$) and less elastic than the selected cross-fitted configuration on Yahoo!R3 and KuaiRand ($\eta = +0.32$ and $+0.33$). We read the audit as consistent with invariance where DR edges are used and informative where IPS edges are used, but of low power in between, and it should be reported with these controls.

9. Discussion and implications

Summary of findings. The repeated-walk bias is real and exactly removable (RQ1). The guarantees hold with nuisances independent of the log and fail when the degree weights come from the log, which is how our experiments fit them (RQ2). The correction leaves accuracy unchanged on four datasets, and it improves score values only under oracle conditions (RQ3). The thinning audit has low power, and the selected configurations are not invariant on two of four datasets (RQ4).

9.1. Theoretical implications

First, propagation-level debiasing must consider the propagation operator, not only the edges it propagates: multiplying unbiased edge estimates does not give an unbiased multi-hop score. The tools that remove the bias (idempotent reduction and Möbius inversion over walk-index coincidences) extend to polynomial filters of odd degree on binary outcomes built from inverse-propensity or DR edge estimates, as far as Theorem 3 goes; the reduction itself is classical, and the contribution is its exact form for graph recommendation. Second, a guarantee stated for fixed nuisances is a statement about the whole pipeline, not about the estimator: cross-fitting the

propensity and the imputation preserves it only if the degree weights and the score constants are also independent of the log (Table 4), a condition that standard debiased graph pipelines do not meet. Third, unbiasedness for the full-exposure target does not imply better rankings: on unexposed candidates the uncorrected DR operator carries a positive multiple of the informative imputation, and the target itself contains the label of the pair it scores (it ranks the semi-synthetic candidates perfectly, nDCG@20 of 1.0, whereas the estimators reach 0.37–0.38), so matching it on observed pairs is not the same as ranking well.

9.2. Practical implications

The corrected and the uncorrected DR operators are equivalent in accuracy within 0.005 nDCG on KuaiRec and identical on two datasets, so a practitioner who needs only rankings loses nothing by using the cheaper uncorrected operator. Where score values are aggregated, for instance into an estimate of the full-exposure utility of an allocation, the uncorrected estimate is biased by 1.8 to 15 times the target over all pairs (Table 3) and the corrected one is unbiased under the conditions of RQ2; this needs nuisances independent of the log, such as DRUP-split, and true propensities, since with estimated propensities all operators overstate utility by 100–540%. The exposure-capped allocation, which compares scores across users, loses 79–84% of the achievable utility under every estimator, so the correction does not help there. On KuaiRec, training-free propagation over the DR graph is a strong, cheap baseline (nDCG@20 0.638, no gradient training, about 50 minutes of nuisance fitting and 3 s per configuration) against which trained debiased graph recommenders should be compared; two thirds of its gain over

the imputation alone comes from propagating the imputation. A thinning audit can probe exposure invariance, but it needs a negative control and a configuration whose scores are not dominated by the imputation. Finally, the sample split that satisfies the assumptions costs up to 0.06 nDCG on sparse logs, which is the price of the guarantee. The properties that follow from linearity (exact attributions, certificates, a private item operator, capped allocation) are in the Supplementary Material, together with their weak results: private release is below popularity on Coat and Yahoo!R3 and for $\epsilon \leq 4$ on KuaiRec, and at 200 fake profiles a DR or DRUP target reaches 39% of KuaiRec users against 10% under IPS.

9.3. Limitations

The analysis is static: it assumes a single logging policy and fixed potential outcomes, whereas recommenders are retrained on their own feedback (Chaney et al., 2018; Hazrati and Ricci, 2022; Perdomo et al., 2020). Proposition 4 covers dependence within a user’s history with known conditional propensities, not dependence across users, and the idempotent correction needs binary outcomes. The main experiments cross-fit the nuisances and compute degree weights and score constants from the log, so the theorems do not cover them (Remark 1). The oracle conditions of the score-level result (true propensities, fixed weights) are not met by any real pipeline. The training-free operators searched 384 to 6,300 configurations against 6 to 108 for the trained models (equal-budget results are in Supplementary Table S2); the KuaiRec margin of 0.004 over DR-JL and MRDR is smaller than their seed spread and not significant under the Nadeau–Bengio test. Yahoo!R3 and KuaiRand select IPS edges, so they do not test the DR half

of the method. The propensities shipped with Coat were estimated with the uniform sample that we use for validation and testing (Schnabel et al., 2016), a possible leak. We did not run LLM-based recommenders (Section 7.2), StableDR, DICE, CausE or AutoDebias, and the largest catalogue has about 10^4 items; the dense implementation does not scale to 10^5 or more. The target F^* is a normative choice that other stakeholders may not share. Removing the dependence of expected scores on exposure does not make a recommender fair to item providers: DRUP’s lists are more concentrated than those of the logged graph (Gini 0.60 against 0.34 on Coat; 1.2% against 15% of the candidate items in the KuaiRec top-20), so a system retrained on its own output could entrench a small set of items. The data are public and anonymised, and no human subjects are involved.

10. Conclusion

Inverse-propensity and doubly robust graph propagation are biased beyond one hop because walks that repeat an edge multiply its estimate with itself. We gave the exact bias and removed it with an idempotent correction, DRUP, that is exact for any odd number of hops and, at three hops, costs about as much as the uncorrected operator. The guarantees hold when the nuisances do not depend on the log, for instance with a sample split, and fail when degree weights are computed from the log. On four benchmarks the correction does not change accuracy against up to 26 tuned baselines, and it improves score values only with true propensities and fixed weights. Future work includes a protocol with log-independent weights that shows a benefit on real logs, guarantees under adaptive logging, audits with controls that

discriminate, and a comparison with text-based recommenders on a dataset that supports it.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support code development and to polish the writing. All ideas and results are the author’s own. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Table 1: DRUP and representative related methods. “loss”: the correction acts on the training objective; “output”: it acts on the predictions; “learned”: learned aggregation weights; “_”: not applicable. Properties that follow from linearity (exact attributions, certificates, joint DP) hold for every linear operator, including linear LightGCN, GF-CF and DR adjacency, and are not listed.

Method	debiases propagation	unbiased beyond 1 hop	doubly robust	exposure invariance
LightGCN, NGCF, UltraGCN, SGL, SimGCL	no	no	no	no
linear LightGCN, EASE, GF-CF, BSPM	no	no	no	no
IPS, SNIPS, DR, MRDR, StableDR	loss	–	DR variants	no
MACR, PDA, DICE	output	–	no	no
NAVIP	edge weights	no	no	no
r-AdjNorm, APDA, CAGED, DPAA	learned	no	no	no
DR adjacency (uncorrected)	DR edges	no	edge-wise	no*
DRUP-split (ours; assumptions hold)	DR edges	any odd K	yes [†]	in expectation [†]
DRUP, cross-fitted (ours; as run)	DR edges	no [‡]	no [‡]	no [‡]

*Over-corrects (Theorem 6) with nuisances independent of the log; as invariant as DRUP in our real-data audits, which cannot tell them apart. [†]Guaranteed under Assumptions 1–2 (the sample split) and unclipped propensities. [‡]No guarantee: nuisances are cross-fitted on the same log and degree weights and score constants are computed from it (Remark 1); in simulation the correction then no longer reduces bias, and the audited configurations are not invariant on Yahoo! R3 and KuaiRand. The cross-fitted operator is what the accuracy tables report; its accuracy equals that of uncorrected DR adjacency.

Table 2: Datasets. “Log”: the biased training data, with its density. “Test”: the unbiased data. “Candidates”: unbiased test items per user that are ranked (items never logged for that user). “Evaluated users”: users with at least one positive candidate.

Dataset	Users / items	Log (density)	O means	Test	Candidates	Evaluated users
Coat	290 / 300	6,960 (8.0%)	Self-rated	16 random ratings per user	14.7	225
Yahoo! R3	15,400 / 1,000	311,704 (2.0%)	Self-rated	10 random songs, 5,400 users	10	2,315
KuaiRand	10,000 / 7,583	511,863 (0.68%)	Feed impression	random impressions	43.6	8,347
KuaiRec	7,176 / 10,728	10.3M (13.4%)	Impression	fully observed $1,411 \times 3,327$	3,314	1,410

Table 3: Monte-Carlo check of Theorems 1, 3, 5 and 6 and Corollary 2. Relative $|\text{bias}|$ is the mean absolute bias divided by the mean target, over all entries and conditionally on unexposed candidates (against the conditional target of Corollary 2). Kendall τ is the mean within-user rank agreement of one realisation with the full-exposure target on unexposed candidates. η is the exposure elasticity (separate model, 3,000 replications).

Estimator	rel. $ \text{bias} $, $K = 3$		rel. $ \text{bias} $, $K = 5$		Kendall τ	elasticity
	all	unexposed	all	unexposed	($K = 3$)	η
Logged graph	–	–	–	–	–	+1.777
IPS adjacency	2.335	0.029	14.641	4.667	0.024	+0.007
IPS + walk correction	0.028	0.029	0.060	0.059	0.024	+0.007
DR adjacency	1.756	1.137	8.350	5.053	0.199	-0.187
DRUP	0.019	0.019	0.039	0.040	0.122	+0.005

Table 4: Monte-Carlo check of the nuisance protocol (Remark 1): relative $|\text{bias}|$ of the three-hop term against the full-exposure target with the same weights C , over all entries, with true propensities. Nuisances are fitted on an independent log (Assumption 2), cross-fitted over ten folds of pairs of the same log (the protocol of the main experiments), or fitted on a random 50% or 20% of the pairs of the same log (sample split). Model: users $\times$ items and mean exposures per user; the last model has three times larger propensities. The last two columns give the bias of the full score $s = s_1/c_1 + s_3/c_3$ with constants computed from the scores or from the imputed graph (2,000–3,000 replications).

Model	Nuisances	IPS edges		DR edges			
		uncorr.	DRUP	uncorr.	DRUP	$s, \text{data } c$	$s, \hat{Y} c$
40 $\times$ 60, 11	independent log	2.15	0.10	1.90	0.09	0.35	0.11
	cross-fitted (protocol)	0.39	0.64	0.61	0.80	0.62	0.88
	cross-fitted, W -degrees	3.12	0.81	4.11	0.73	0.49	2.69
	sample split, 50%	0.60	0.42	0.55	0.42	0.35	0.63
	sample split, 20%	1.09	0.17	0.90	0.17	0.29	0.24
60 $\times$ 120, 23	independent log	0.98	0.07	0.82	0.06	0.30	0.07
	cross-fitted (protocol)	0.31	0.52	0.42	0.61	0.42	0.58
	cross-fitted, W -degrees	1.51	0.69	1.80	0.58	0.40	1.09
	sample split, 50%	0.32	0.39	0.31	0.39	0.35	0.52
	sample split, 20%	0.54	0.18	0.47	0.18	0.26	0.26
60 $\times$ 120, 60	independent log	0.15	0.01	0.09	0.01	0.18	0.02
	cross-fitted (protocol)	0.11	0.17	0.12	0.18	0.20	0.12
	cross-fitted, W -degrees	0.17	0.18	0.11	0.12	0.18	0.08
	sample split, 50%	0.09	0.13	0.10	0.13	0.35	0.33
	sample split, 20%	0.09	0.07	0.07	0.07	0.23	0.14

Table 5: Semi-synthetic KuaiRec (30 draws of the log). RMSE T : root mean squared error of the three-hop term over all pairs, relative to the root mean square of the target (only for fixed C , where estimator and target use the same weights). Utility error: relative error (mean $\pm$ standard error over draws) of the estimated full-exposure utility $\sum_x s$ of a fixed random and a fixed popular allocation of 20 unexposed items per user. Regret: loss of full-exposure utility of the exposure-capped allocation computed from the estimated scores, relative to the one computed from F^* . nDCG@20 against Y on unexposed items. True propensities; estimated ones and a sparser log are in the Supplementary Material.

p	Nuisances	C	Estimator	RMSE T	utility error (%)			
					random	popular	regret	nDCG@20
true	independent log	fixed	IPS	22.8	+17.8±4.0	+10.3±0.4	0.795	0.097
			IPS+WC	2.9	+0.4±1.2	+0.4±0.3	0.795	0.097
			DR	28.2	+23.0±4.9	+17.8±0.5	0.791	0.097
			DRUP	6.3	-1.8±1.4	-0.0±0.4	0.795	0.107
true	independent log	$\hat{Y}$	IPS	–	-44.0±1.0	-41.3±0.2	–	0.120
			IPS+WC	–	-48.4±0.8	-44.4±0.2	–	0.120
			DR	–	-44.0±1.0	-40.4±0.2	–	0.377
			DRUP	–	-49.3±0.8	-44.6±0.2	–	0.370
true	cross-fitted	fixed	IPS	22.8	+17.8±4.0	+10.3±0.4	0.795	0.097
			IPS+WC	2.9	+0.4±1.2	+0.4±0.3	0.795	0.097
			DR	35.3	+28.6±4.7	+18.9±0.5	0.821	0.051
			DRUP	8.5	-0.1±1.6	+0.8±0.4	0.819	0.068
true	cross-fitted	$\hat{Y}$	IPS	–	-54.0±0.7	-46.5±0.1	–	0.191
			IPS+WC	–	-56.3±0.6	-48.5±0.1	–	0.191
			DR	–	-53.7±0.9	-45.4±0.1	–	0.320
			DRUP	–	-57.4±0.8	-48.8±0.1	–	0.322
true	sample split	fixed	IPS	20.6	+53.2±3.9	+55.9±0.6	0.822	0.093
			IPS+WC	2.5	+37.9±1.2	+47.2±0.6	0.821	0.091
			DR	103.6	+63.3±5.0	+63.7±0.8	0.840	0.030
			DRUP	8.8	+37.9±1.5	+47.0±0.6	0.837	0.048
true	sample split	$\hat{Y}$	IPS	–	+0.8±1.9	-10.4±0.6	–	0.154
			IPS+WC	–	-10.7±1.3	-17.1±0.5	–	0.150
			DR	–	+0.9±1.8	-10.3±0.6	–	0.274
			DRUP	–	-10.8±1.2	-17.2±0.5	–	0.272

Five hops (independent log, true p , fixed C): relative |bias| of T_5 , DR 3.46, DRUP 0.32.

Table 6: Accuracy on unbiased test data (mean $\pm$ std over splits; for the sample-split rows, mean $\pm$ std over five draws of the split-off pairs). Best in bold, second underlined. “–”: not run (five hops only where the three-hop DR operators are competitive). Every method is tuned per split on the same validation data; Table S1 lists the number of configurations. Only the sample-split variants satisfy Assumption 2.

Method	Coat nDCG@5	Yahoo! R3 nDCG@5	KuaiRand nDCG@10	KuaiRec nDCG@20
<i>Non-personalised</i>				
Popularity	0.5367 $\pm$ 0.0095	0.5331 $\pm$ 0.0059	0.4204 $\pm$ 0.0019	0.5766 $\pm$ 0.0023
Imputation only	0.5532 $\pm$ 0.0097	0.5162 $\pm$ 0.0044	0.3637 $\pm$ 0.0015	0.6194 $\pm$ 0.0018
<i>Trained, pointwise loss</i>				
MF	0.4780 $\pm$ 0.0168	0.5087 $\pm$ 0.0075	0.3705 $\pm$ 0.0008	0.6221 $\pm$ 0.0036
IPS-MF	0.4713 $\pm$ 0.0160	0.5125 $\pm$ 0.0059	–	0.6252 $\pm$ 0.0018
DR-MF	0.5237 $\pm$ 0.0111	0.5093 $\pm$ 0.0042	0.3608 $\pm$ 0.0067	0.6219 $\pm$ 0.0025
DR-JL	0.4811 $\pm$ 0.0164	0.5225 $\pm$ 0.0043	0.3759 $\pm$ 0.0021	0.6332 $\pm$ 0.0026
MRDR	0.4869 $\pm$ 0.0187	0.5248 $\pm$ 0.0059	–	0.6328 $\pm$ 0.0020
iALS	0.5812$\pm$0.0101	0.6625 $\pm$ 0.0043	0.4528 $\pm$ 0.0017	0.6133 $\pm$ 0.0027
LightGCN	0.5389 $\pm$ 0.0107	0.5947 $\pm$ 0.0028	0.4061 $\pm$ 0.0014	0.6271 $\pm$ 0.0023
DR-LightGCN	0.5724 $\pm$ 0.0205	0.5528 $\pm$ 0.0091	–	0.6212 $\pm$ 0.0071
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.5174 $\pm$ 0.0206	0.6461 $\pm$ 0.0042	0.4475 $\pm$ 0.0027	0.5963 $\pm$ 0.0028
PDA	0.5181 $\pm$ 0.0138	0.6514 $\pm$ 0.0051	–	0.5815 $\pm$ 0.0027
MACR	0.5418 $\pm$ 0.0134	0.6221 $\pm$ 0.0038	0.4311 $\pm$ 0.0019	0.5816 $\pm$ 0.0033
LightGCN (BPR)	0.5683 $\pm$ 0.0132	<u>0.6708$\pm$0.0061</u>	<u>0.4569$\pm$0.0010</u>	0.6085 $\pm$ 0.0028
r-AdjNorm	0.5722 $\pm$ 0.0134	0.6742$\pm$0.0047	0.4590$\pm$0.0024	0.6083 $\pm$ 0.0031
NAVIP	0.5664 $\pm$ 0.0163	0.6701 $\pm$ 0.0039	0.4565 $\pm$ 0.0017	0.6186 $\pm$ 0.0024
SimGCL	0.5528 $\pm$ 0.0141	0.6348 $\pm$ 0.0048	0.4196 $\pm$ 0.0022	0.5637 $\pm$ 0.0031
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.5571 $\pm$ 0.0100	0.6585 $\pm$ 0.0058	0.4468 $\pm$ 0.0018	0.6036 $\pm$ 0.0028
EASE	0.5401 $\pm$ 0.0138	0.6478 $\pm$ 0.0061	0.4350 $\pm$ 0.0019	0.5213 $\pm$ 0.0032
GF-CF	0.5513 $\pm$ 0.0123	0.6552 $\pm$ 0.0039	0.4502 $\pm$ 0.0016	0.6141 $\pm$ 0.0025
BSPM	<u>0.5727$\pm$0.0090</u>	0.6507 $\pm$ 0.0047	0.4498 $\pm$ 0.0025	0.6143 $\pm$ 0.0026
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6132 $\pm$ 0.0030
IPS + walk correction	0.5510 $\pm$ 0.0224	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6132 $\pm$ 0.0030
propagation over $\hat{Y}$ only (no residual)	0.5533 $\pm$ 0.0171	0.5280 $\pm$ 0.0043	–	0.6312 $\pm$ 0.0016
EASE on DR graph	0.5564 $\pm$ 0.0141	0.6072 $\pm$ 0.0033	0.4030 $\pm$ 0.0011	0.6025 $\pm$ 0.0022
GF-CF on DR graph	0.5608 $\pm$ 0.0083	0.6312 $\pm$ 0.0051	0.4468 $\pm$ 0.0012	0.6372 $\pm$ 0.0020
BSPM on DR graph	0.5624 $\pm$ 0.0112	0.6262 $\pm$ 0.0071	0.4483 $\pm$ 0.0016	0.6354 $\pm$ 0.0016
DR adjacency	0.5459 $\pm$ 0.0244	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	<u>0.6378$\pm$0.0020</u>
DRUP	0.5467 $\pm$ 0.0235	0.6567 $\pm$ 0.0063	0.4492 $\pm$ 0.0013	0.6378 $\pm$ 0.0019
DR adjacency, 5 hops	0.5524 $\pm$ 0.0245	–	–	0.6379$\pm$0.0011
DRUP, 5 hops	0.5451 $\pm$ 0.0204	–	–	0.6376 $\pm$ 0.0011
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.5447 $\pm$ 0.0129	0.5952 $\pm$ 0.0037	0.4400 $\pm$ 0.0007	0.6322 $\pm$ 0.0035
DRUP-split	0.5443 $\pm$ 0.0112	0.5947 $\pm$ 0.0041	0.4400 $\pm$ 0.0007	0.6321 $\pm$ 0.0035

Table 7: User-level paired comparisons: mean difference of the test metric over users with a 95% interval, and the t -test p -value, Holm-adjusted over all comparisons run for the dataset (DRUP, DR adjacency and DRUP-split, and on Coat and KuaiRec also five-hop DRUP, each against every other method: 124 on Coat, 87 on Yahoo! R3, 75 on KuaiRand, 120 on KuaiRec). Users with a positive candidate who are in the test part of at least one split (225, 2,310, 8,321 and 1,410); sample-split rows use the first draw. Selected rows; all comparisons are in the result files.

Comparison	Coat			Yahoo! R3			KuaiRand			KuaiRec		
	Δ [95% CI]	p_{Holm}		Δ [95% CI]	p_{Holm}		Δ [95% CI]	p_{Holm}		Δ [95% CI]	p_{Holm}	
DRUP vs DR adjacency (the correction)	+0.0014 [-0.002, +0.005]	1.000		+0.0000 [+0.000, +0.000]	1.000		+0.0000 [+0.000, +0.000]	1.000		-0.0001 [-0.000, +0.000]	1.000	
DRUP vs DR adjacency, both 5 hops	-0.0081 [-0.015, -0.001]	1.000		-	-		-	-		-0.0003 [-0.001, -0.000]	0.170	
DRUP-split vs DR adjacency, sample split	-0.0033 [-0.010, +0.003]	1.000		-0.0012 [-0.004, +0.001]	1.000		+0.0002 [-0.000, +0.001]	1.000		-0.0001 [-0.000, +0.000]	1.000	
DRUP-split vs DRUP (cost of the split)	+0.0087 [-0.014, +0.032]	1.000		-0.0659 [-0.077, -0.055]	< 10 ⁻³		-0.0092 [-0.012, -0.006]	< 10 ⁻³		-0.0035 [-0.005, -0.002]	0.004	
DRUP vs linear LightGCN, logged graph	-0.0123 [-0.051, +0.027]	1.000		-0.0022 [-0.005, +0.001]	1.000		+0.0025 [+0.001, +0.005]	0.267		+0.0346 [+0.030, +0.039]	< 10 ⁻³	
DRUP vs imputation only	-0.0082 [-0.017, +0.001]	1.000		+0.1411 [+0.125, +0.157]	< 10 ⁻³		+0.0861 [+0.081, +0.091]	< 10 ⁻³		+0.0179 [+0.016, +0.020]	< 10 ⁻³	
DRUP vs GF-CF	-0.0070 [-0.046, +0.032]	1.000		+0.0005 [-0.005, +0.006]	1.000		-0.0013 [-0.004, +0.001]	1.000		+0.0243 [+0.021, +0.028]	< 10 ⁻³	
DRUP vs BSPM	-0.0295 [-0.065, +0.006]	1.000		+0.0051 [-0.001, +0.011]	1.000		-0.0005 [-0.003, +0.002]	1.000		+0.0241 [+0.021, +0.027]	< 10 ⁻³	
DRUP vs GF-CF on DR graph	-0.0143 [-0.023, -0.006]	0.106		+0.0247 [+0.018, +0.032]	< 10 ⁻³		+0.0025 [+0.001, +0.004]	0.943		+0.0004 [-0.001, +0.002]	1.000	
DRUP vs iALS	-0.0372 [-0.071, -0.003]	1.000		-0.0065 [-0.014, +0.001]	1.000		-0.0031 [-0.006, +0.000]	1.000		+0.0249 [+0.021, +0.028]	< 10 ⁻³	
DRUP vs DR-JL	+0.0658 [+0.039, +0.093]	< 10 ⁻³		+0.1350 [+0.120, +0.150]	< 10 ⁻³		+0.0739 [+0.069, +0.079]	< 10 ⁻³		+0.0044 [+0.002, +0.007]	0.003	
DRUP vs LightGCN (pointwise)	+0.0073 [-0.014, +0.028]	1.000		+0.0616 [+0.048, +0.075]	< 10 ⁻³		+0.0443 [+0.040, +0.049]	< 10 ⁻³		+0.0103 [+0.007, +0.013]	< 10 ⁻³	
DRUP vs LightGCN (BPR)	-0.0252 [-0.058, +0.008]	1.000		-0.0146 [-0.023, -0.006]	0.021		-0.0081 [-0.011, -0.005]	< 10 ⁻³		+0.0298 [+0.026, +0.033]	< 10 ⁻³	
DRUP vs r-AdjNorm	-0.0290 [-0.062, +0.004]	1.000		-0.0187 [-0.027, -0.010]	< 10 ⁻³		-0.0101 [-0.013, -0.007]	< 10 ⁻³		+0.0296 [+0.026, +0.033]	< 10 ⁻³	
DRUP vs NAVIP	-0.0236 [-0.056, +0.009]	1.000		-0.0138 [-0.022, -0.005]	0.029		-0.0077 [-0.011, -0.004]	< 10 ⁻³		+0.0194 [+0.017, +0.022]	< 10 ⁻³	
DRUP vs SimGCL	-0.0077 [-0.033, +0.018]	1.000		+0.0219 [+0.012, +0.032]	< 10 ⁻³		+0.0302 [+0.026, +0.034]	< 10 ⁻³		+0.0742 [+0.070, +0.079]	< 10 ⁻³	
DRUP-split vs iALS	-0.0285 [-0.062, +0.005]	1.000		-0.0724 [-0.084, -0.061]	< 10 ⁻³		-0.0123 [-0.016, -0.008]	< 10 ⁻³		+0.0214 [+0.018, +0.025]	< 10 ⁻³	
DRUP-split vs DR-JL	+0.0745 [+0.043, +0.106]	< 10 ⁻³		+0.0691 [+0.057, +0.081]	< 10 ⁻³		+0.0647 [+0.060, +0.069]	< 10 ⁻³		+0.0009 [-0.002, +0.003]	1.000	
DRUP-split vs LightGCN (pointwise)	+0.0160 [-0.010, +0.042]	1.000		-0.0043 [-0.019, +0.010]	1.000		+0.0351 [+0.031, +0.040]	< 10 ⁻³		+0.0068 [+0.004, +0.010]	< 10 ⁻³	
DRUP-split vs r-AdjNorm	-0.0203 [-0.052, +0.012]	1.000		-0.0846 [-0.096, -0.073]	< 10 ⁻³		-0.0193 [-0.023, -0.016]	< 10 ⁻³		+0.0261 [+0.023, +0.030]	< 10 ⁻³	

Table 8: Real-data intervention $do(p \leftarrow p/2)$: the logged exposures of a random half of the items are thinned with probability 1/2 (three to five draws). Rank shift: change of the treated items’ mean within-user rank relative to control items (0: invariant). η : elasticity of the mean score of treated relative to control candidates (1: exposure-proportional, 0: invariant). Frozen: imputation and degree weights C are kept, the propensities are updated to the known $p/2$; re-fitted: all nuisances are re-fitted on the thinned log, with the known propensity change or with propensities re-estimated on it. One configuration per operator (the one selected most often), all test data.

Data	Operator	frozen nuisances		re-fitted, known $p/2$		re-fitted, re-estimated $\hat{p}$	
		rank shift	η	rank shift	η	rank shift	η
Coat	logged graph	-0.0670	+1.031	-0.0585	+0.748	–	–
	IPS adjacency	-0.0652	+0.899	-0.0651	+0.900	–	–
	DR adjacency	+0.0021	-0.005	+0.0001	-0.019	–	–
	DRUP	+0.0036	-0.008	+0.0015	-0.022	–	–
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 94%						
Yahoo! R3	logged graph	-0.1611	+1.002	-0.0943	+0.508	-0.0943	+0.508
	IPS adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DR adjacency	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	DRUP	-0.1504	+0.959	-0.0864	+0.481	-0.0873	+0.485
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 95%						
KuaiRand	logged graph	-0.0826	+0.997	-0.0679	+0.711	-0.0679	+0.711
	IPS adjacency	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	DR adjacency	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	DRUP	-0.0678	+0.639	-0.0681	+0.651	-0.0680	+0.653
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 57%						
KuaiRec	logged graph	-0.2408	+0.997	-0.1339	+0.500	-0.1339	+0.500
	IPS adjacency	-0.0462	+0.137	-0.0466	+0.141	-0.0657	+0.218
	DR adjacency	+0.0061	-0.013	+0.0001	-0.003	+0.0017	-0.006
	DRUP	+0.0057	-0.012	-0.0004	-0.002	+0.0011	-0.005
	treated logged pairs below the DRUP clip after $do(p \leftarrow p/2)$: 16%						

Table 9: Controls for the thinning audit (same intervention as Table 8, η averaged over draws). Negative control: the thinned log is scored with the propensities of the original log, so that the estimator is misspecified by a factor of two on treated pairs. Bottom: Spearman correlation of the scores with the imputation $\hat{Y}$ over all candidate pairs.

η	Coat	Yahoo! R3	KuaiRand	KuaiRec
popularity (re-fitted)	+0.993	+0.998	+1.000	+1.000
imputation only (re-fitted)	-0.010	-0.007	-0.014	-0.160
IPS adjacency, propensities not updated	+1.019	+1.002	+0.997	+0.997
DR adjacency, propensities not updated	-0.004	+1.002	+0.997	-0.087
DRUP, propensities not updated	-0.008	+1.002	+0.997	-0.088
DRUP-split, $\hat{Y}$ and C frozen	+0.022	+0.319	+0.328	-0.003
DRUP-split, re-fitted (known change)	+0.018	+0.319	+0.325	-0.007
DRUP-split, re-fitted (estimated)	–	+0.327	+0.331	-0.013
rank corr. with $\hat{Y}$: logged graph	0.30	0.27	0.36	0.36
rank corr. with $\hat{Y}$: DR adjacency	0.99	0.27	0.37	0.55
rank corr. with $\hat{Y}$: DRUP	0.98	0.27	0.37	0.54
rank corr. with $\hat{Y}$: DRUP-split	0.76	0.58	0.55	0.42